

Document 3 — Project Design Phase

Incident Lifecycle Automation in ServiceNow

Project Title: Incident Lifecycle Automation in ServiceNow

College Project Portfolio

Date: August 23, 2026

1. System Overview

The proposed system is designed to automate and standardize the IT Incident Management lifecycle. The solution ensures that every reported IT issue follows a predictable path from initial logging to final resolution and knowledge capture. By leveraging the ServiceNow platform, the system centralizes support activities into a professional digital workflow.

Major ServiceNow components included in the design:

- **Incident Management:** Core module for record-keeping and workflow.
- **Service Operations Workspace:** Modern agent interface for efficient handling.
- **Agent Assist & Knowledge Management:** AI-driven solution recommendations.
- **Change Management:** Integration for infrastructure modifications.
- **Service Level Agreements (SLAs):** Automated performance tracking.
- **CMDB Integration:** Linking incidents to Services and Configuration Items (CIs).

2. Solution Architecture

The high-level architecture focuses on a seamless flow of information between the End User, the Service Desk, and the backend technical support teams. The design utilizes a centralized ServiceNow instance as the system of record.

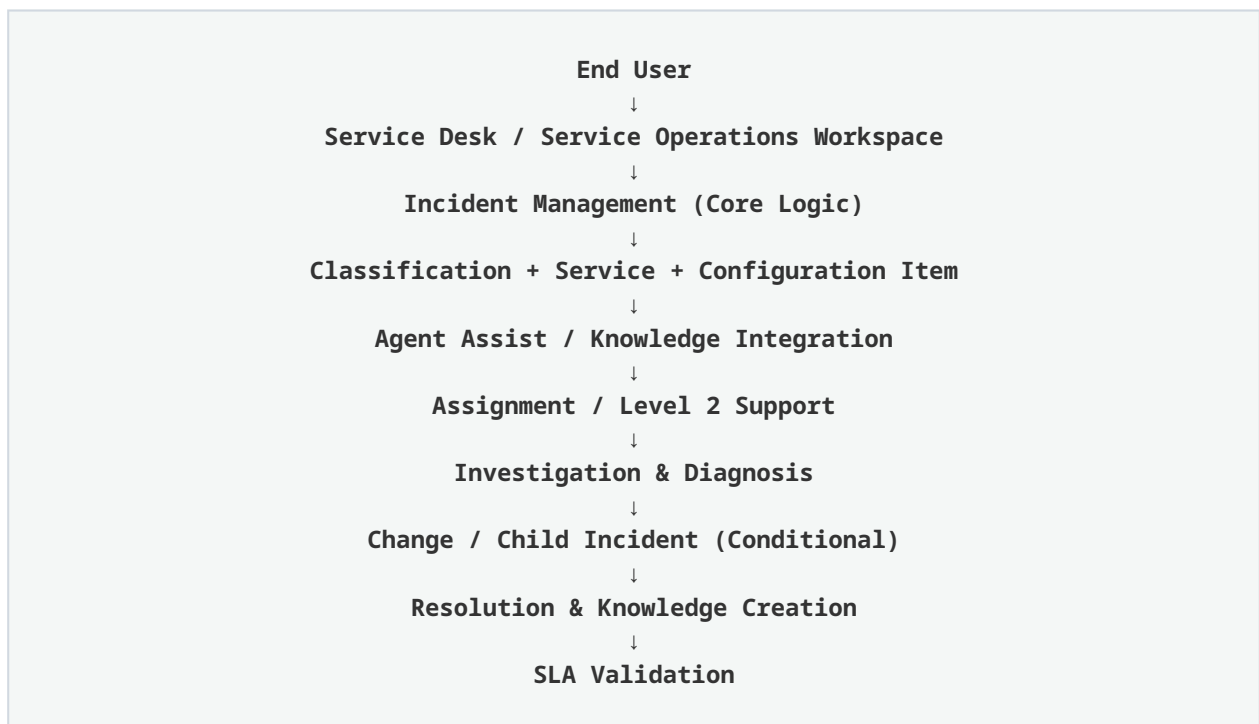


Figure 1: High-level architecture of the proposed ServiceNow Incident Management solution.

3. Incident Lifecycle Workflow

The designed workflow ensures governance at every stage:

- **Incident Creation:** Capturing caller details, short description, and intake channel.
- **Classification:** Categorising by Service, Category, Subcategory, and Urgency.
- **Knowledge Search:** Using Agent Assist to identify existing solutions.
- **Assignment:** Routing to the Service Desk or Level 2 groups.
- **Investigation:** Documentation of findings and technical steps.
- **Change if Required:** Association with Emergency Changes for service restoration. *(Note: Workflow designed for Change integration, though verification screenshots were omitted per project documentation).*
- **Resolution:** Final documentation of cause and resolution code.
- **Knowledge Creation:** Converting successful resolutions into reusable articles.
- **SLA Validation:** Monitoring Response and Resolution timers via Task SLAs.

4. Service and Service Offering Design

The system uses a hierarchical service structure to improve classification and reporting visibility. The design links specific technical offerings to broader business services.

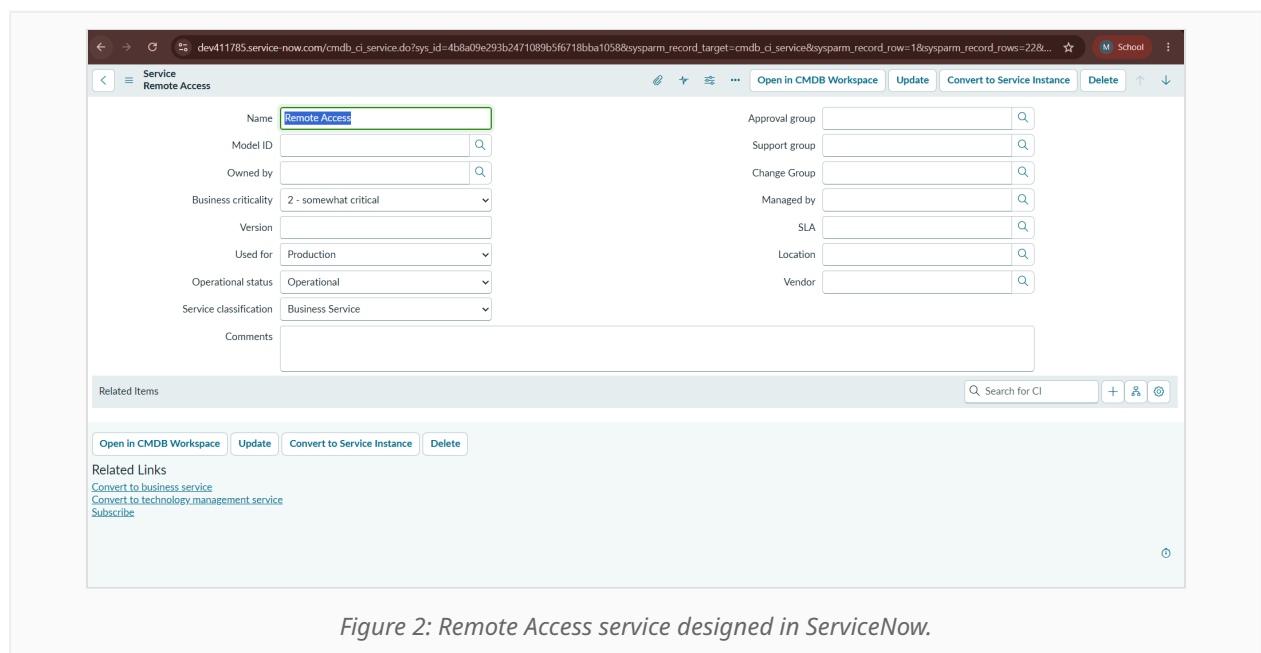


Figure 2: Remote Access service designed in ServiceNow.

The screenshot shows the 'Offering - Corporate VPN' form in ServiceNow. The form is divided into several sections. The top section contains fields for Name (Corporate VPN), Model ID, Owned by, Parent (Remote Access), Used for (Production), Operational status (Operational), and Vendor. The middle section contains fields for Approval group, Support group, Change group, Managed by, Version, Business criticality (2 - somewhat critical), SLA, and Location. The bottom section contains text areas for Short Description, Description, and Comments. At the very bottom, there are 'Related Items' and 'Contract' fields.

Figure 3: Corporate VPN service offering associated with Remote Access.

The relationship **Remote Access** → **Corporate VPN** allows the system to automatically track the health of specific offerings while reporting on the overall status of Remote Access capabilities.

5. Incident Design

The incident record is designed to capture structured data that drives automation. By selecting the correct Service and Category, the system can provide contextual help and ensure the correct SLA is applied.

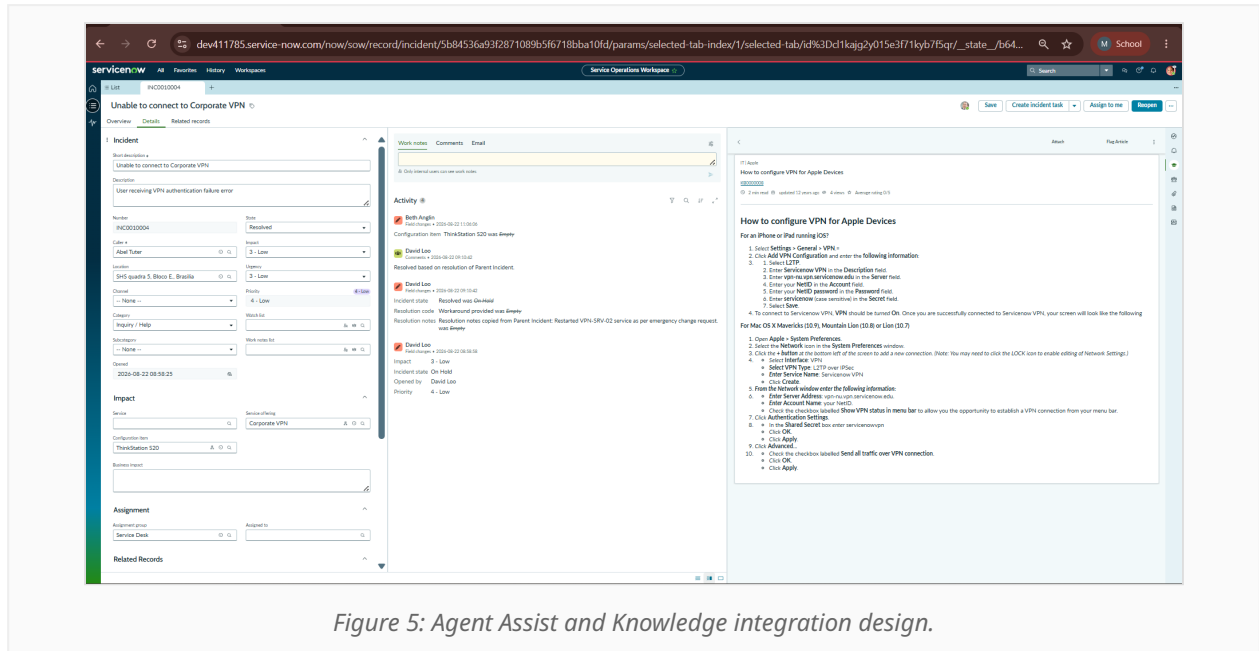
The screenshot shows the 'Incident' form in ServiceNow. The form is divided into several sections. The top section contains fields for Incident number (INC0010004), Date, Category (Network), Subcategory (VPN), Channel (Phone), Priority (Low), Impact (Low), Service (Corporate VPN), and Assignment group (Service Desk). The middle section contains a 'Work notes' section with a list of activities. The bottom section contains a 'Record Information' section with fields for Caller, Caller phone number, and Assigned to.

Figure 4: Designed incident classification fields in ServiceNow.

Fields such as Channel (Phone), Category (Network), and Subcategory (VPN) are mandatory to ensure data quality for future reporting.

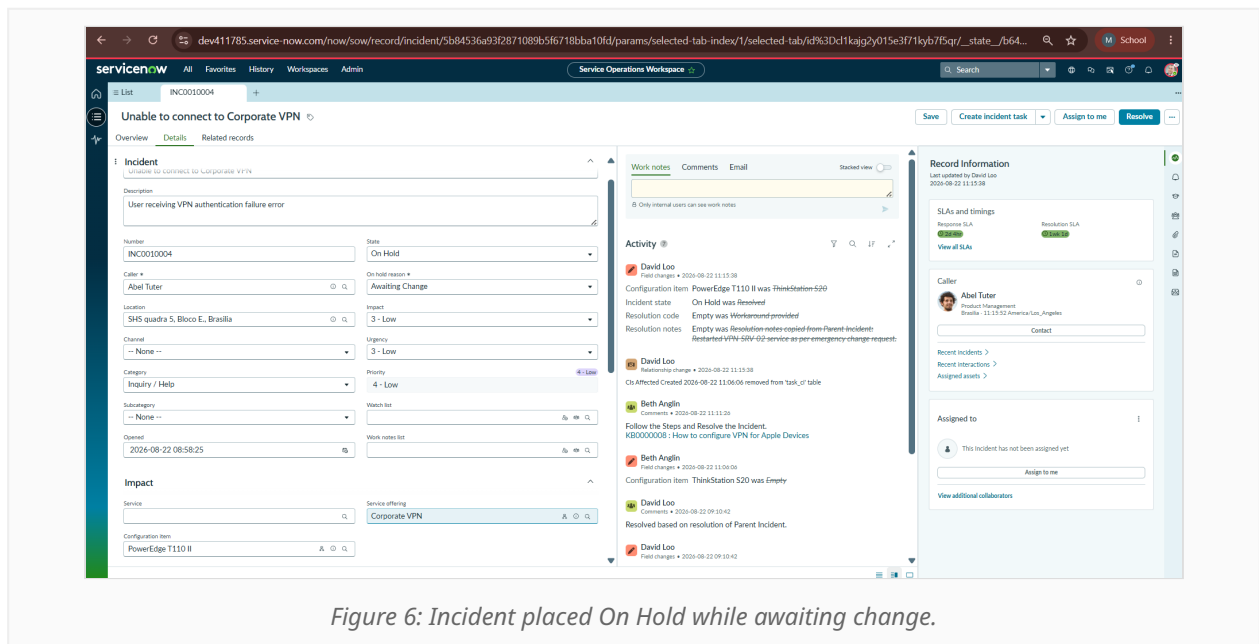
6. Knowledge Integration Design

Knowledge integration is designed to empower agents using **Agent Assist**. The process involves searching for relevant keywords, reviewing suggested articles, and attaching them to the incident. This reduces resolution time and ensures consistent troubleshooting steps are followed.



7. Change Integration Design

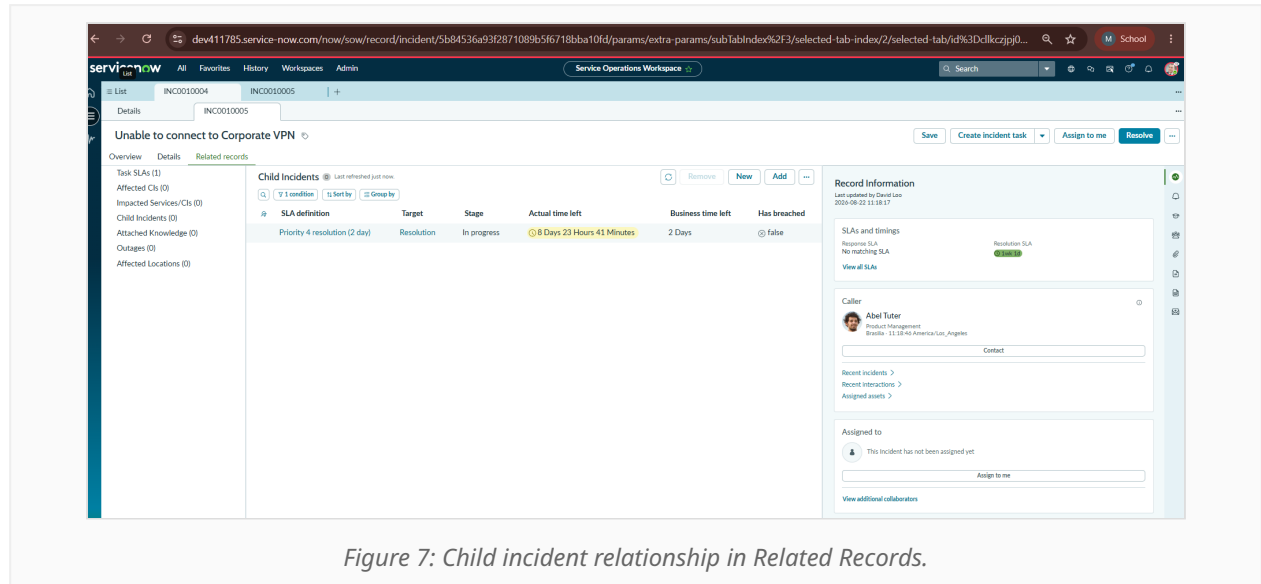
In cases where service restoration requires a structural fix, the design allows for Change Management integration. The incident can be placed in an **On Hold** state with the reason **Awaiting Change**, pausing the resolution SLA while the change is being processed.



Requirement Note: While the Change Request verification screenshot was not captured in this project instance, the design includes the placeholder and state transition logic required for integration.

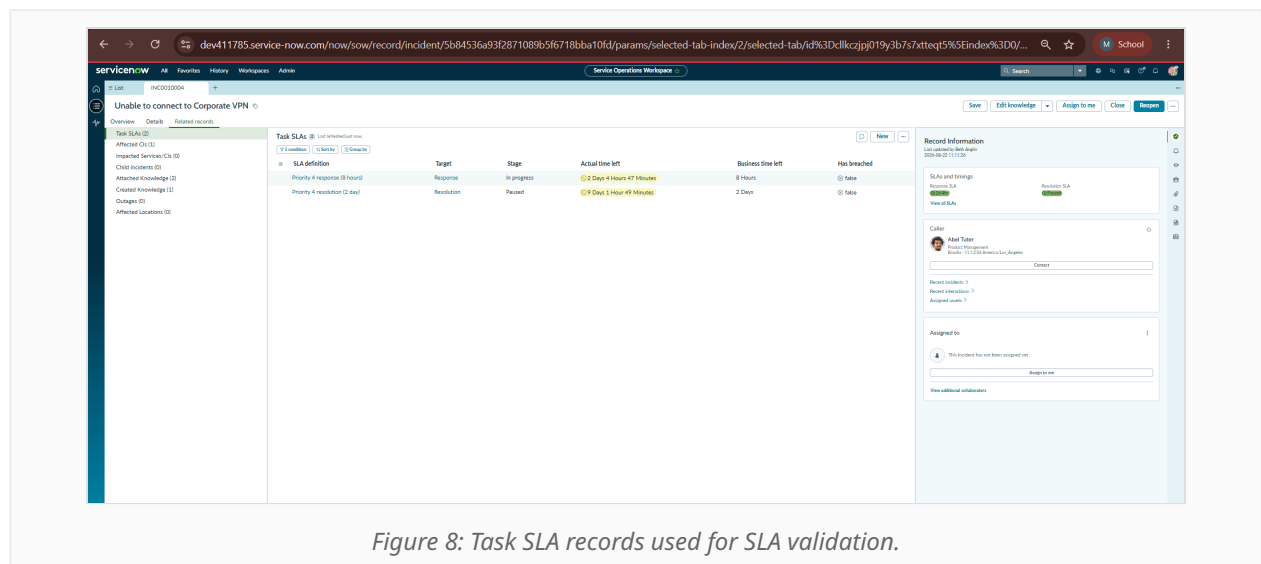
8. Child Incident Design

The design supports the linking of multiple incidents via a Parent-Child relationship. This is critical for managing widespread issues (e.g., VPN outage affecting multiple users) by allowing the Service Desk to update many records through a single parent incident.



9. SLA Design

Service Level Agreements are monitored using the **Task SLA** related list. Each incident tracks "Response" and "Resolution" targets. This provides visibility into the business time remaining before a breach occurs, ensuring support teams prioritise tickets effectively.



10. Role-Based Design

The design assigns clear responsibilities to each stakeholder group:

- **Service Desk Agents:** Intake, classification, and initial triage.

- **Level 2 Support:** Advanced diagnosis and infrastructure investigation.
- **Incident Manager:** Governance and SLA compliance monitoring.
- **Knowledge Manager:** Standardisation of resolution articles.

11. Expected Workflow Summary

The final design represents an end-to-end lifecycle where the incident is identified (Step 1-2), classified (Step 4), supported by knowledge (Step 5-6), escalated or held for change if needed (Step 10), and finally resolved with validated SLAs (Step 8) and documented knowledge.

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